

ENGINEERING REBOOT 2026

Week 02 – Day 10 Field Manual v3.5

PUBLICATION EDITION (CORRECTED)

PHASE 0 – MISSION BRIEF

Objective

Integrate functions, conditionals, loops, and user input into a single working application.

Core Concepts

- Functions
- Loops
- Conditionals
- User Input
- Program Structure
- Integration of Concepts

Deliverable Inventory

1. calculator_application.py
2. Notes.md
3. Journal.md
4. Evidence Package

PHASE 1 – MINI PROJECT DESIGN REVIEW

Filename

calculator_application.py

Application Requirements

- Display menu options
- Accept user selection
- Request numeric inputs
- Perform calculations
- Display results
- Allow repeated execution

- Exit only when user chooses Exit

Required Menu

===== CALCULATOR =====

- 1 - Add
- 2 - Subtract
- 3 - Multiply
- 4 - Divide
- 5 - Exit

Required Functions

display_menu()
add_numbers()
subtract_numbers()
multiply_numbers()
divide_numbers()

Validation Matrix

Expected Startup Display:

===== CALCULATOR =====

- 1 - Add
- 2 - Subtract
- 3 - Multiply
- 4 - Divide
- 5 - Exit

Expected Result:

- Menu displays correctly
- User can select an option

Evidence Checkpoint

calculator_application-Menu.png

PHASE 2 – ADDITION OPERATION

Input: 10, 15

Expected Output: Result: 25

Evidence: calculator_application-Addition.png

PHASE 3 – SUBTRACTION OPERATION

Input: 25, 10

Expected Output: Result: 15

Evidence: calculator_application-Subtraction.png

PHASE 4 – MULTIPLICATION OPERATION

Input: 6, 7

Expected Output: Result: 42

Evidence: calculator_application-Multiplication.png

PHASE 5 – DIVISION OPERATION

Input: 100, 4

Expected Output: Result: 25.0

Additional Validation:

Input: 10, 0

Expected Output: Cannot divide by zero.

Evidence:

calculator_application-Division.png

calculator_application-DivideByZero.png

PHASE 6 – PROGRAM LOOP

Display Menu → Select Operation → Perform Calculation → Display Result → Return To Menu

Continue until Exit is selected.

Evidence: calculator_application-LoopExecution.png

PHASE 7 – NOTES

Create:

Week-02/Notes/Day-10/Notes.md

Required Sections:

- Concepts Learned
- Key Commands
- Key Observations
- Lessons Learned
- Questions For Future Study

PHASE 8 – JOURNAL

Create:

Week-02/Journal/Day-10/Journal.md

Required Sections:

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

PHASE 9 – GIT OPERATIONS

git status

git add .

git commit -m "Complete Day 10 calculator application"

git push

git fetch

git status

git branch -vv

Preferred End State:

Your branch is up to date with 'origin/main'

nothing to commit, working tree clean

Evidence: Git-Operations.png

END-OF-DAY SUCCESS CRITERIA

- calculator_application.py completed
- All calculator operations validated
- Divide-by-zero handling implemented
- Program loop implemented
- Notes completed
- Journal completed
- Git workflow completed
- Repository synchronized

DAY 11 PREVIEW

Topic: Lists and Collections

Expected Deliverables:

- list_basics.py
- shopping_list.py
- list_statistics.py

Core Concepts:

- lists
- indexing
- iteration
- list methods
- collection processing